Management Information Systems

A management information system is an **Information System** that provides accurate, timely and organized information for decision makers to make decisions. It provides the required information to the managers to perform their jobs.

Information System

An information system is a collection of hardware, software, data, people and procedures that work together to produce quality information. This system stores, retrieves, transforms and distributes information in an organization.

Chapter 1

Information Systems in Global Business Today

Management Information Systems
Chapter 1 Information Systems in Global Business Today

The Role of Information Systems in Business Today

- **Information** meaningful **data** (facts)
- Societies mainly depend on the information
- Information support to achieve desired objectives
    - Planning, organizing, decision making, …
- **IS** process (handle) the data to be useful information
    - Through H/W, S/W, telecommunication, DB
- IS computer systems related to process data and provide required information
- ATMs, airline reservation systems, course reservation systems

BUSINESS PROCESS

- Business process : tasks, rules, behaviors that been developed to produce business results.
- eg.
  - Developing new product
  - Creating market plan
  - Hiring an employee
- Considered source of competitive strength
- IS automate many business process

DEFINITIONS

- Data
- Information
- System
- Information system
- Business
- Business process

ROLE OF IS IN BUSINESS TODAY

- IS transform (improve) business today. How?

1. Creating globalization opportunities: Internet reduced costs of operating, on global scale for Customers and firms, Using foreign markets, easily replicate service such as Google and ebay

2. Helps the emerging digital firm: The business relationships in digital firm are digitally enabled and mediated and It's core **business processes** are accomplished through digital networks.

   (Time shifting, space shifting)

FIRMS INVEST HEAVILY IN IS TO ACHIEVE 6 STRATEGIC BUSINESS OBJECTIVES (WHY FIRMS USE IS?).

1. Operational excellence
2. New products, services, and business models
3. Customer and supplier intimacy
4. Improved decision making
5. Competitive advantage
6. Survival

1. OPERATIONAL EXCELLENCE:

- IS Improvement of efficiency of operation to attain higher profitability

  - IT tool to achieve greater efficiency and productivity

  - Examble: Wal-Mart’s RetailLink ***system*** links suppliers to stores for superior replenishment system

2. NEW PRODUCTS, SERVICES, AND BUSINESS MODELS

- Business model: describes how company produces, delivers, and sells product or service to create wealth
- Information systems and technology a major enabling tool for new products, services, business models
  - E.g. Apple's iPod, iTunes and Netflix's Internet-based DVD rentals

3. CUSTOMER AND SUPPLIER INTIMACY

- Serving **customers** well leads to customers returning, which raises revenues and profits
  - E.g. High-end hotels that use computers to track customer preferences and use to monitor and customize environment
- Intimacy with **suppliers** allows them to provide vital inputs, which lowers costs
  - E.g. J.C.Penney's information system which links sales records to contract manufacturer

4. IMPROVED DECISION-MAKING

- Without accurate information:
  - Managers must use forecasts, best guesses, luck
  - Leads to:
    - Overproduction, underproduction of goods and services
    - Misallocation of resources
    - Poor response times
  - Poor outcomes raise costs, lose customers
- IS provide real-time data for making decisions
- E.g. Verizon's Web-based digital dashboard to provide managers with real-time data on customer complaints, network performance, line outages, etc.

5. COMPETITIVE ADVANTAGE

- Achieve higher sales and profit through using IS by:
  - Doing things better
  - Charging less for superior products
  - real time Responding
- Using the internet is competitive advantage
- E.g. Dell: Consistent profitability over 25 years; Dell remains one of the most efficient producer of PCs in world.
- But Dell has lost some of its advantages to fast followers-- HP

6. Survival

- Information technologies are necessity of doing business
- May be:
  - Industry-level changes, e.g. Citibank's introduction of ATMs
  - Governmental regulations requiring record-keeping
    - E.g. Toxic Substances Control Act, Sarbannes-Oxley Act

IS, DATA, INFORMATION

- **IS**
  - Set of interrelated components
  - Collect, process, store, and distribute information

  →By computers and software as a tool
  - Support decision making, coordination, control, problem analysis and create new product
  - Provide solutions to challenges in business Env.
- **Information vs. data**
  - Data are streams of raw facts
  - Information is data shaped into meaningful form

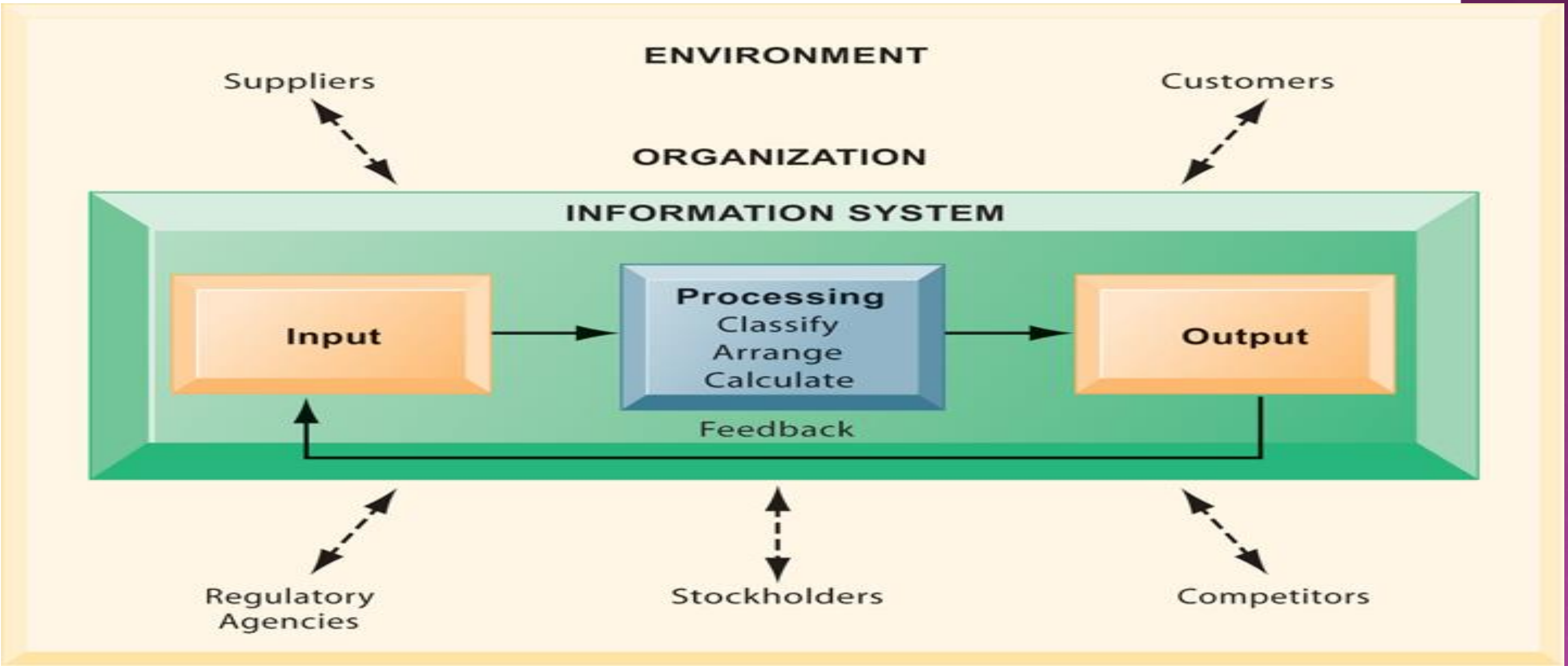
IS: INPUT, PROCESS, OUTPUT, FEEDBACK

- **IS:** activities produce required information
  - **Input**: Captures raw data from organization or external environment
  - **Processing**: Converts data into meaningful form
  - **Output**: Transfers processed information to people or activities that use it
  - **Feedback**: Output returned to appropriate members of organization to help evaluate or correct input stage

Management Information Systems
Chapter 1 Information Systems in Global Business Today

Perspectives on Information Systems

Functions of an Information System



An information system contains information about an organization and its surrounding environment. Three basic activities—input, processing, and output—produce the information organizations need. Feedback is output returned to appropriate people or activities in the organization to evaluate and refine the input. Environmental actors, such as customers, suppliers, competitors, stockholders, and regulatory agencies, interact with the organization and its information systems.

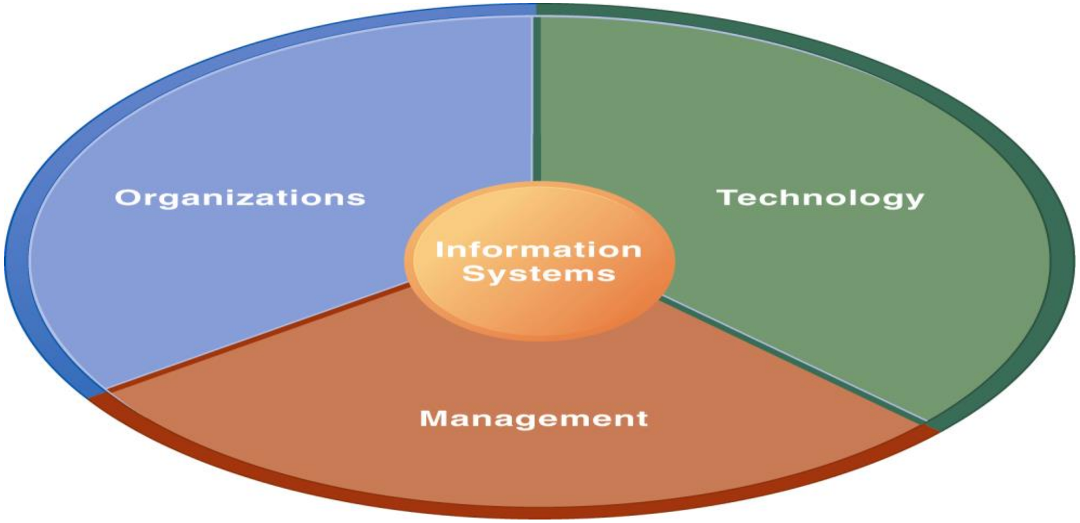
Figure 1-4

COMPUTER/COMPUTER PROGRAM VS. INFORMATION SYSTEM

- Computers and software are technical foundation and tools to store and process information
- similar to the material and tools used to build a house
- Cannot produce required information to a particular organization

DIMENSIONS (BOARDERS) OF IS

- Understanding of IS dimensions is IS literacy
- Where computer literacy is focus on primarily on knowledge of IT
- MIS try to achieve this boarders, deals with behavioral and technical issues surrounding development, use and impact of IS in the firm.



1. ORGANIZATIONAL DIMENSION OF IS

- structure: different levels and specialties
  - hierarchy of authority, responsibility: Senior Middle Operational management, Knowledge service Data workers
- business process: Organization coordinate its work through its hierarchy and business process
- Culture : ways of doing things, part is embedded in IS.

Levels in a Firm

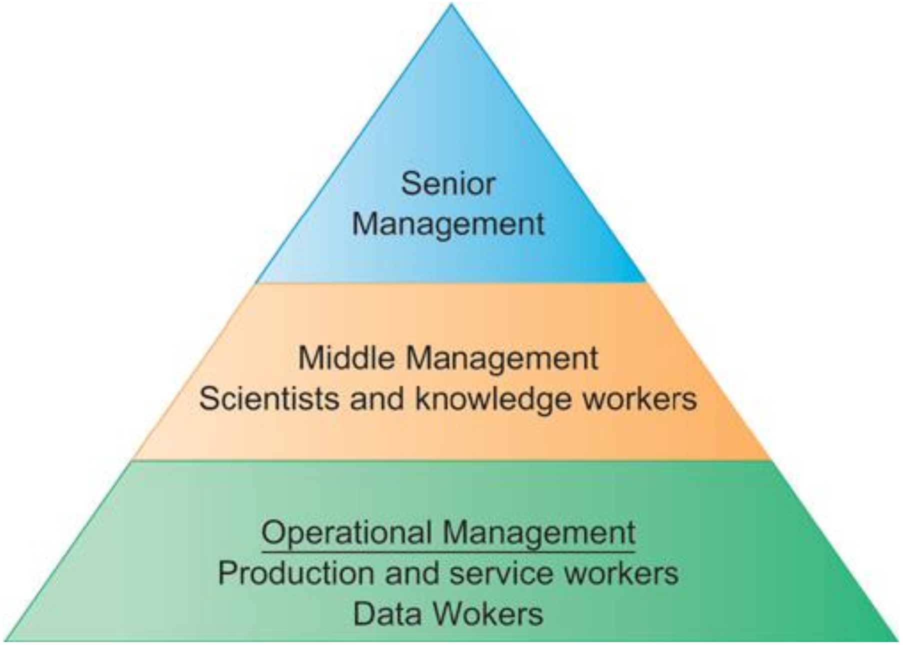


Figure 1-6

2. MANAGEMENT DIMENSION

- Make decisions, formulate action plan and solve organizational problem
- Managers set organizational strategy for responding to business challenges
- In addition, managers must act creatively:
  - Creation of new products and services
  - Occasionally re-creating the organization

3. TECHNOLOGY DIMENSION

- IT is:
  - Hardware: physical component
  - Software: instruction control Hardware
  - Data management technology
  - Network and telecommunications technology
  - Network: Hardware + Software link computers
  - Internet: Network to provide services
  - WWW: service to store retrieve information
- IT infrastructure: platform that the firm can built on its IS

BUSINESS PERSPECTIVE ON IS:

- IS instrument for creating value to firms
- Investments in IS result in superior returns:
  - Increases productivity and revenue
- IS provides information that helps managers making better decisions and improve the execution of business process
- Value of IS

THERE VARIATION IN RETURNS ON INFORMATION TECHNOLOGY INVESTMENT

- Investing in information technology does not guarantee good returns
- Considerable variation in the returns firms receive from systems investments

- Factors that case The **variation** in **Returns** :
  - Adopting right business model according (suite) to new technology
  - complementary investments (business processes, models, management behavior and culture)

COMPLEMENTARY ASSETS:

- Assets required to derive value from a primary investment
- Firms supporting their technology investments with investment in complementary assets receive superior returns
- E.g.: invest in technology <u>and</u> the people to make it work properly

CONTEMPORARY APPROACHES TO INFORMATION SYSTEMS

TECHNICAL APPROACH:

- Emphasizes mathematically based models
  - Computer science theories of commutation ,data storage
  - management science: models of DM and practices
  - operations research: optimizing selected parameters of org.

BEHAVIORAL APPROACH

- : Behavioral issues such strategic business integration, implementation...
    - Psychology: how decision makers use formal information
    - Economics: how IS change the control and cost structures
    - Sociology: how system affect individuals and groups

MANAGEMENT INFORMATION SYSTEMS MIS

- Use of computer-based information systems in business firms
- Combines work of CS, management, and operating research toward developing system solutions to real word problems
- Concerned with behavioral issues of development, use and impact of IS
- **main actors :** Suppliers of hardware and software, Business firms, Managers and employees, Firm’s environment (legal, social, cultural context)

Chapter 2

Global E-Business: How Businesses Use Information Systems

DEFINITIONS

- **Business processes:**
  - Workflows of material, information, knowledge
  - Sets of activities, steps
  - May be tied to functional area or be cross-functional
- **Businesses**: Can be seen as collection of business processes
- Business processes may be assets or liabilities

INFORMATION TECHNOLOGY AND BUSINESS PROCESSES

- **Information technology enhances business processes in two main ways:**
  - **Increasing efficiency of existing processes**
    - Automating steps that were manual
  - **Enabling entirely new processes that are capable of transforming the businesses**
    - Change flow of information
    - Replace sequential steps with parallel steps
    - Eliminate delays in decision making

FUNCTIONAL BUSINESS PROCESSES

- **Examples :**
  - **Manufacturing and production**
    - Assembling the product
  - **Sales and marketing**
    - Identifying customers
  - **Finance and accounting**
    - Creating financial statements
  - **Human resources**
    - Hiring employees

SALES AND MARKETING SYSTEMS

- Functional concerns include:
  - Sales management, customer identification market research, advertising and promotion, pricing, new products
- Examples of systems:
  - Order processing (operational level)
  - Pricing analysis (middle mgmt)
  - Sales trend forecasting (senior mgmt)

MANUFACTURING AND PRODUCTION SYSTEMS

- Functional concerns include:
  - Managing production facilities, production goals, production materials, and scheduling
- Examples of systems:
  - Machine control (operational mgmt)
  - Production planning (middle mgmt)
  - Facilities location (senior mgmt)

FINANCE AND ACCOUNTING SYSTEMS

- Functional concerns include:
  - Managing financial assets (cash, stocks, etc.) and capitalization of firm, and managing firm's financial records
- Examples of systems:
  - Accounts receivable (operational mgmt)
  - Budgeting (middle mgmt)
  - Profit planning (senior mgmt)

HUMAN RESOURCE SYSTEMS

- Functional concerns include:
  - Identifying potential employees, maintaining employee records, creating programs to develop employee talent and skills
- Examples of systems:
  - Training and development (operational mgmt)
  - Compensation analysis (middle mgmt)
  - Human resources planning (senior mgmt)

SYSTEMS FROM A CONSTITUENCY PERSPECTIVE

- Transaction processing systems: supporting operational level employees
- Management information systems and decision-support systems: supporting managers
- Executive support systems: supporting executives

TRANSACTION PROCESSING SYSTEMS

- Perform and record daily routine transactions necessary to conduct business
  - E.g. sales order entry, payroll, shipping
- Allow managers to monitor status of operations and relations with external environment
- Serve operational levels
- Serve predefined, structured goals and decision making

MANAGEMENT INFORMATION SYSTEMS

- Serve middle management
- Provide reports on firm's current performance, based on data from TPS
- Provide answers to routine questions with predefined procedure for answering them
- Typically have little analytic capability

DECISION SUPPORT SYSTEMS

- Serve middle management
- Support nonroutine decision making
  - E.g. What is impact on production schedule if December sales doubled?
- Often use external information as well from TPS and MIS
- Model driven DSS
  - Voyage-estimating systems
- Data driven DSS
  - Intrawest’s marketing analysis systems

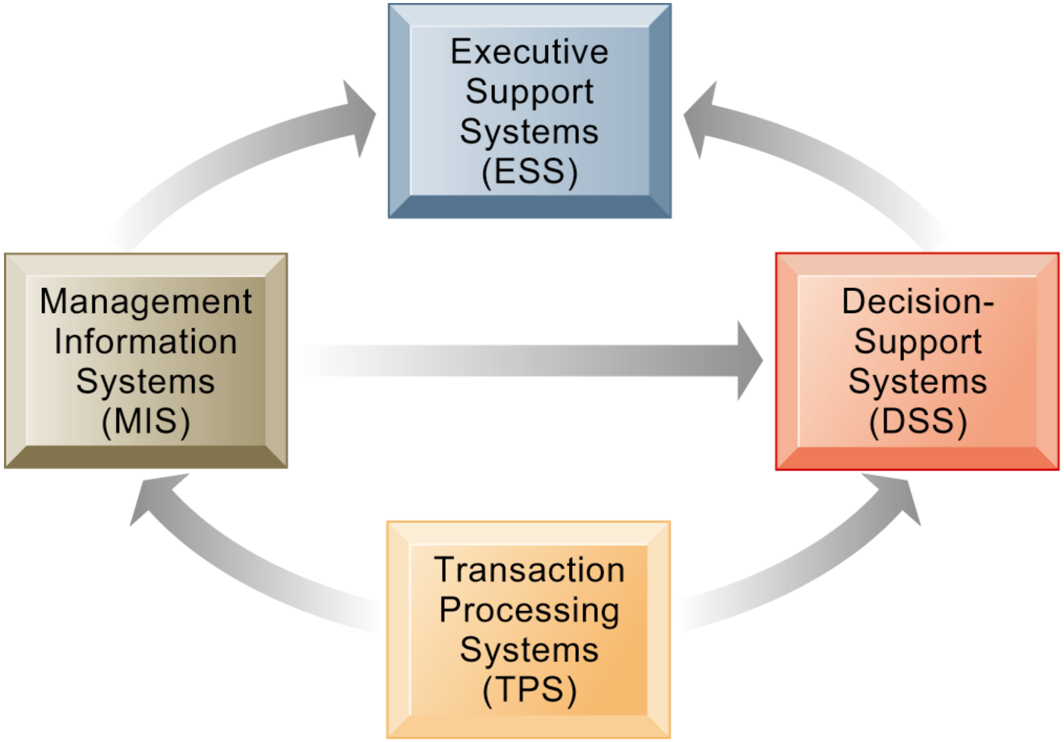
EXECUTIVE SUPPORT SYSTEMS

- Support senior management
- Address nonroutine decisions requiring judgment, evaluation, and insight
- Incorporate data about external events (e.g. new tax laws or competitors) as well as summarized information from internal MIS and DSS
- E.g. ESS that provides minute-to-minute view of firm's financial performance as measured by working capital, accounts receivable, accounts payable, cash flow, and inventory.

RELATIONSHIP OF SYSTEMS TO ONE ANOTHER

- TPS: Major source of data for other systems
- ESS: Recipient of data from lower-level systems
- Data may be exchanged between systems

Interrelationships Among Systems



The various types of systems in the organization have interdependencies. TPS are major producers of information that is required by many other systems in the firm, which, in turn, produce information for other systems. These different types of systems are loosely coupled in most business firms, but increasingly firms are using new technologies to integrate information that resides in many different systems.

Figure 2-10

PROBLEM:

- Different kinds if IS in a firm work together
- The challenge to get them all work together as one corporate system(information integration)
- Solution: applications (systems) that span functional areas, focus on:
- Execute, coordinate business process
- Integrate group of process
- Focus on management of resources and customer service

SOLUTION: ENTERPRISE APPLICATIONS

- Include all levels of management
- Execute business processes across firm
- Span functional areas
- **Types**
  - Enterprise systems (ERP)
  - Supply chain management systems
  - Customer relationship management systems:
  - Knowledge management systems KMS

ENTERPRISE SYSTEMS (ERP)

- Collects data from different firm functions and stores data in single central data storage area to be shared among firm
- Resolves problem of fragmented, redundant data sets and systems
- Enable:
  - Coordination of daily activities
  - Efficient response to customer orders (production, inventory)
  - Provide valuable information for improving management decision making

SUPPLY CHAIN MANAGEMENT SYSTEMS

- Manage firm's relationships with suppliers
- Share information about
  - Orders, production, inventory levels, delivery of products and services
- Goal: Right amount of products to destination with least amount of time and lowest cost
- Inaccurate info: excessive inventories; inaccurate manufacturing plans; missed production schedule; raise cost less satisfaction

CUSTOMER RELATIONSHIP MANAGEMENT SYSTEMS:

- Provide information to coordinate all of the business processes that deal with customers in sales, marketing, and service to optimize revenue, customer satisfaction, and customer retention.

- Integrate firm's customer-related processes and support customer information from multiple communication channels (tel., email, WiFi devices)

KNOWLEDGE MANAGEMENT SYSTEMS KMS

- Knowledge and experience on how to create produce and deliver products and services
- Collect Knowledge and experience and make available whenever and wherever
- KMS Support processes for acquiring, creating, storing, distributing, applying, integrating knowledge, link internal knowledge to external knowledge
- Include enterprise-wide systems for: Managing documents, graphics and other digital knowledge objects; Directories of employees with expertise
-

INTRANETS AND EXTRANETS:

- An **intranet** is a private network contained within an enterprise that is used to securely share company information and computing resources among employees. An **intranet** can also be used to facilitate working in groups and teleconferences.

- An **extranet** is a controlled private network that allows access to partners, vendors and suppliers or an authorized set of customers – normally to a subset of the information accessible from an organization's intranet.

E-BUSINESS, E-GOVERNMENT

- **E-business (Electronic business):**
  - Use of digital technology and Internet to execute major business processes in the enterprise
  - Includes **e-commerce** (electronic commerce):
    - Buying and selling of goods over Internet
- **E-government:**
  - The application of Internet and networking technologies to digitally enable government and public sector agencies' relationships with citizens, businesses, and other arms of government

Chapter 3

Information Systems, Organizations, and Strategy

ORGANIZATION

- An organized group of people with a particular purpose, such as a business or government department.

- An organization, or organisation, is an entity – such as a company, an institution, or an association – comprising one or more people and having a particular purpose.

ORGANIZATION

- IT and organizations influence one another
- Complex relationship influenced by:
  - organization's structure:
  - business processes:
  - Politics:
  - culture:
  - Environment:
  - management decisions:

WHAT IS AN ORGANIZATION?

- **Technical definition:**
  - Stable, formal social structure that takes resources from environment and processes them to produce outputs
  - A formal legal entity with internal rules and procedures, as well as a social structure
- **Behavioral definition:**
  - A collection of rights, privileges, obligations, and responsibilities that is delicately balanced over a period of time through conflict and conflict resolution

FEATURES OF ORGANIZATIONS

- Use of hierarchical structure, authority in decision making
- Hire and promote employees based on qualifications
- Devotion to principle of efficiency (maximize output)
- Other features : Routines and business processes, politics, culture, environments and structures.

ROUTINES AND BUSINESS PROCESSES

- **Routines** (standard operating procedures)
  - Precise rules, procedures, and practices developed to cope with virtually all expected situations
- **Business processes:** Collections of routines
- **Business firm:** Collection of business processes

ORGANIZATIONAL POLITICS, CULTURE AND ENVIRONMENT

- **Organizational politics:** Viewpoints about how resources, rewards and punishments should be distributed ,Politics may defeat plans for an IS
- **Organizational culture:** Encompasses set of assumptions that define goal and product
        - What products the organization should produce
        - How and where it should be produced
        - For whom the products should be produced
    - May be powerful unifying force as well as restraint on change
- **Organizational environments:**
    - Organizations can influence their environments
    - Environments generally change faster than organizations
    - IS can be instrument of environmental scanning, act as a lens

ORGANIZATIONAL STRUCTURE

- Structure is shape, 5 kinds (table 3-1):
  - Entrepreneurial: Small start-up business
  - Machine bureaucracy: Midsize manufacturing firm
  - Divisionalized bureaucracy: Fortune 500 firms
  - Professional bureaucracy: Law firms, school systems, hospitals
  - Adhocracy: Consulting firms

OTHER ORGANIZATIONAL FEATURES

- Have Goals, to be achieved
- Constituencies: serve different groups
- Leadership styles : democratic or authoritarian
- Tasks and technology (routine and nonroutine)

3.2 HOW INFORMATION SYSTEM IMPACTS ORGANIZATIONS AND BUSINESS FIRMS

- Economic impacts
- Organizational and behavioral impacts

ECONOMIC IMPACTS

- IT changes relative costs of capital and costs of information
- reduce number and replace the function of more middle managers,
- reduce the need for other forms of capital (buildings, machinery).
- IT helps firms contract in size: it can reduce transaction costs (the cost of participating in markets); Outsourcing
- IS technology is a factor of production, like capital & labor
- IT affects the cost and quality of information and changes economics of information

TRANSACTION COST THEORY

- Firms seek to economize on cost of participating in market (transaction costs)
- Production vs. transaction cost
- IT lowers market transaction costs for firm, making it worthwhile for firms to transact with other firms rather than grow the number of employees.
- With the Internet, firms mind it more cost effective to use the marketplace and contract for work in a market, rather than hire employees.

AGENCY THEORY

- Firm is nexus of contracts among self-interested parties requiring supervision
- Firms experience agency costs (the cost of managing and supervising) which rise as firm grows
- IT can reduce agency costs, making it possible for firms to grow without adding to the costs of supervising, and without adding employees

ORGANIZATIONAL AND BEHAVIORAL IMPACTS

- **IT flattens organizations**
  - Decision making pushed to lower levels
  - Fewer managers needed (IT enables faster decision making and increases span of control)
- **Postindustrial organizations**
  - Organizations flatten because in postindustrial societies, authority increasingly relies on knowledge and competence rather than formal positions

ORGANIZATIONAL RESISTANCE TO CHANGE

- Information systems become bound up in organizational politics because they influence access to a key resource (the information)
- Information systems potentially change an organization’s structure, culture, politics, and work
- Most common reason for failure of large projects is due to organizational and political resistance to change.
- workers may resist changes that disrupt their routines so IS cannot implemented

How Information Systems Impact Organizations and Business Firms

- **The Internet and organizations**
  - **The Internet increases the accessibility, storage, and distribution of information and knowledge for organizations**
  - **The Internet can greatly lower transaction and agency costs**
    - Example: Large firm delivers internal manuals to employees via a corporate Web site, saving millions of dollars in distribution costs

How Information Systems Impact Organizations and Business Firms

- Central organizational factors to consider when planning a new system:
  - Environment
  - Structure
    - Hierarchy, specialization, routines, business processes
  - Culture and politics
  - Type of organization and style of leadership
  - Main interest groups affected by system; attitudes of end users
  - Tasks, decisions, and business processes the system will assist

Using Information Systems to Achieve Competitive Advantage

- **Why do some firms become leaders in their industry?**
- **Michael Porter's competitive forces model**
  - **Provides general view of firm, its competitors, and environment**
  - **Five competitive forces shape fate of firm**
    1. Traditional competitors
    2. New market entrants
    3. Substitute products and services
    4. Customers
    5. Suppliers

Using Information Systems to Achieve Competitive Advantage

PORTER'S COMPETITIVE FORCES MODEL

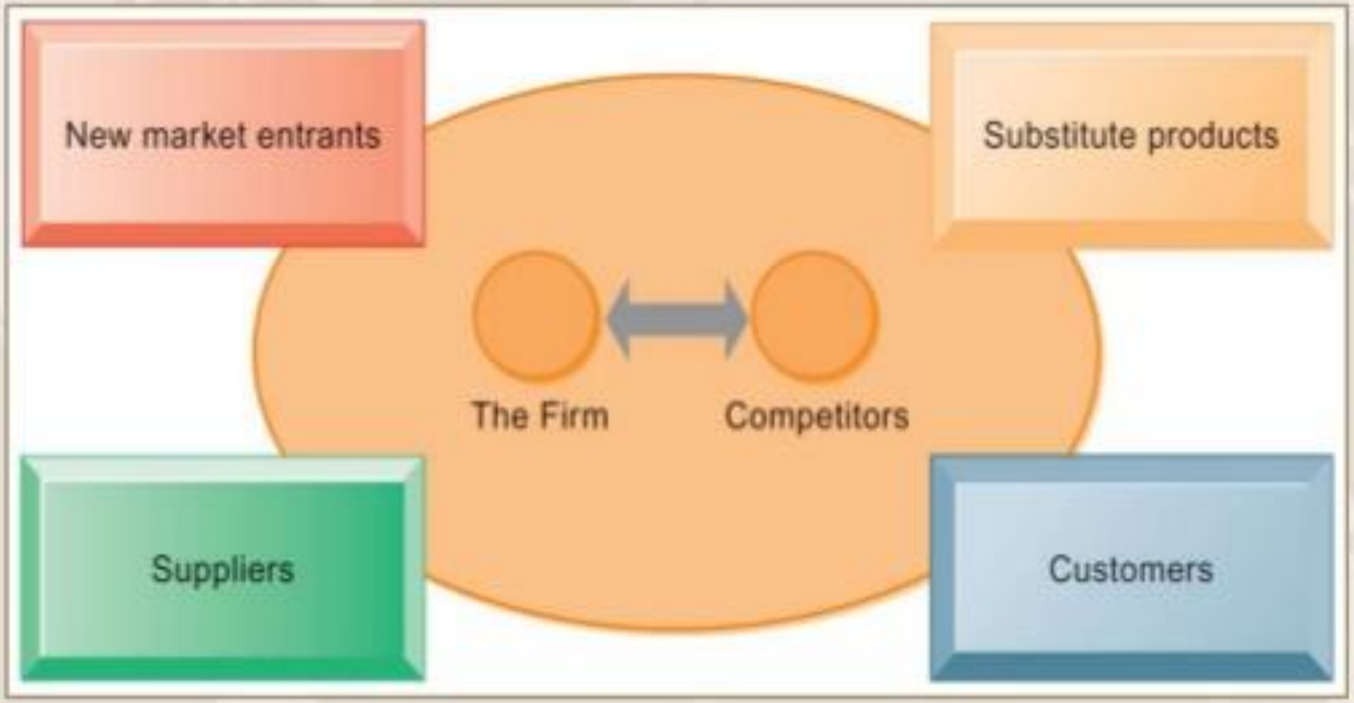


FIGURE 3-10 In Porter's competitive forces model, the strategic position of the firm and its strategies are determined not only by competition with its traditional direct competitors but also by four other forces in the industry's environment: new market entrants, substitute products, customers, and suppliers.

Using Information Systems to Achieve Competitive Advantage

- Traditional competitors
  - All firms share market space with competitors who are continuously devising new products, services, efficiencies, switching costs
- New market entrants
  - Some industries have high barriers to entry, e.g. computer chip business
  - New companies have new equipment, younger workers, but little brand recognition

Using Information Systems to Achieve Competitive Advantage

- Substitute products and services
  - Substitutes customers might use if your prices become too high, e.g. iTunes substitutes for CDs
- Customers
  - Can customers easily switch to competitor's products? Can they force businesses to compete on price alone in transparent marketplace?
- Suppliers
  - Market power of suppliers when firm cannot raise prices as fast as suppliers

Using Information Systems to Achieve Competitive Advantage

- Four generic strategies for dealing with competitive forces, enabled by using IT
  - Low-cost leadership
  - Product differentiation
  - Focus on market niche
  - Strengthen customer and supplier intimacy

Using Information Systems to Achieve Competitive Advantage

- Low-cost leadership
  - Produce products and services at a lower price than competitors while enhancing quality and level of service
  - Examples: Wal-Mart
- Product differentiation
  - Enable new products or services, greatly change customer convenience and experience
  - Examples: Google, Nike, Apple

Using Information Systems to Achieve Competitive Advantage

- Focus on market niche
  - Use information systems to enable a focused strategy on a single market niche; specialize
  - Example: Hilton Hotels
- Strengthen customer and supplier intimacy
  - Use information systems to develop strong ties and loyalty with customers and suppliers; increase switching costs
  - Example: Netflix, Amazon

Using Information Systems to Achieve Competitive Advantage

HOW MUCH DO CREDIT CARD COMPANIES KNOW ABOUT YOU?

Read the Interactive Session and discuss the following questions

- **What competitive strategy are the credit card companies pursuing? How do information systems support that strategy?**
- **What are the business benefits of analyzing customer purchase data and constructing behavioral profiles?**
- **Are these practices by credit card companies ethical? Are they an invasion of privacy? Why or why not?**

Using Information Systems to Achieve Competitive Advantage

- The Internet's impact on competitive advantage
  - Transformation, destruction, threat to some industries
    - E.g. travel agency, printed encyclopedia, newspaper
  - Competitive forces still at work, but rivalry more intense
  - Universal standards allow new rivals, entrants to market
  - New opportunities for building brands and loyal customer bases

MANAGEMENT INFORMATION SYSTEM

MIS (308)

BBA 6th Semester

Using Information Systems to Achieve Competitive Advantage

- Business value chain model
  - Views firm as series of activities that add value to products or services
  - Highlights activities where competitive strategies can best be applied
    - Primary activities vs. support activities
  - At each stage, determine how information systems can improve operational efficiency and improve customer and supplier intimacy
  - Utilize benchmarking, industry best practices

Using Information Systems to Achieve Competitive Advantage

THE VALUE CHAIN MODEL

This figure provides examples of systems for both primary and support activities of a firm and of its value partners that can add a margin of value to a firm's products or services.

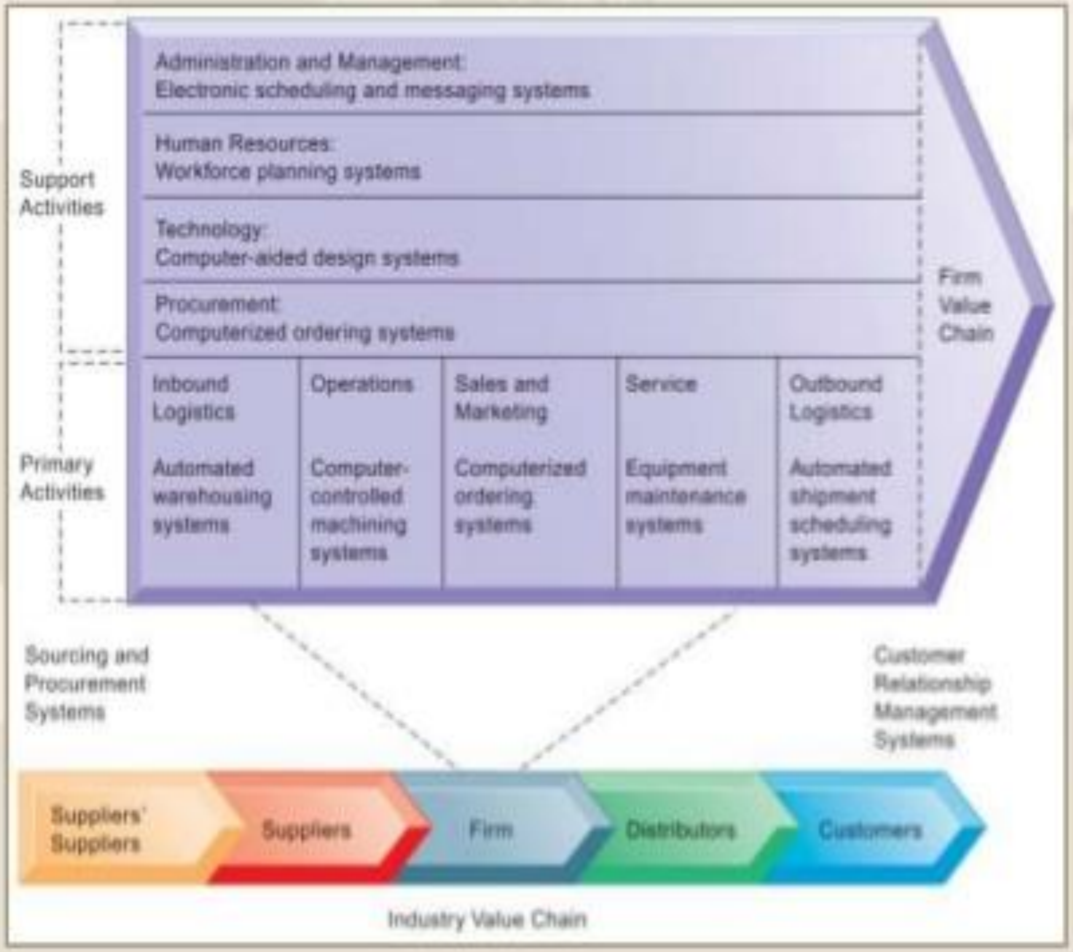


FIGURE 3-11

Using Information Systems to Achieve Competitive Advantage

- Value web:
  - Collection of independent firms using highly synchronized IT to coordinate value chains to produce product or service collectively
  - More customer driven, less linear operation than traditional value chain

Using Information Systems to Achieve Competitive Advantage

- **Information systems can improve overall performance of business units by promoting synergies and core competencies**
  - **Synergies**
    - When output of some units used as inputs to others, or organizations pool markets and expertise
    - Example: merger of Bank of NY and JPMorgan Chase
    - Purchase of YouTube by Google

Using Information Systems to Achieve Competitive Advantage

- **Core competencies**
  - **Activity for which firm is world-class leader**
  - **Relies on knowledge, experience, and sharing this across business units**
  - **Example: Procter & Gamble's intranet and directory of subject matter experts**

Using Information Systems to Achieve Competitive Advantage

- **Network-based strategies**
  - **Take advantage of firm's abilities to network with each other**
  - **Include use of:**
    - Network economics
    - Virtual company model
    - Business ecosystems

Using Information Systems to Achieve Competitive Advantage

- **Traditional economics: Law of diminishing returns**
  - The more any given resource is applied to production, the lower the marginal gain in output, until a point is reached where the additional inputs produce no additional outputs
- **Network economics:**
  - Marginal cost of adding new participant almost zero, with much greater marginal gain
  - Value of community grows with size
  - Value of software grows as installed customer base grows

Using Information Systems to Achieve Competitive Advantage

- Virtual company strategy
  - Virtual company uses networks to ally with other companies to create and distribute products without being limited by traditional organizational boundaries or physical locations
  - E.g. Li & Fung manages production, shipment of garments for major fashion companies, outsourcing all work to over 7,500 suppliers

Using Information Systems to Achieve Competitive Advantage

- **Business ecosystems**
  - **Industry sets of firms providing related services and products**
    - Microsoft platform used by thousands of firms
    - Wal-Mart's order entry and inventory management
  - **Keystone firms:** Dominate ecosystem and create platform used by other firms
  - **Niche firms:** Rely on platform developed by keystone firm
  - **Individual firms can consider how IT will help them become profitable niche players in larger ecosystems**

Using Information Systems to Achieve Competitive Advantage

AN ECOSYSTEM STRATEGIC MODEL

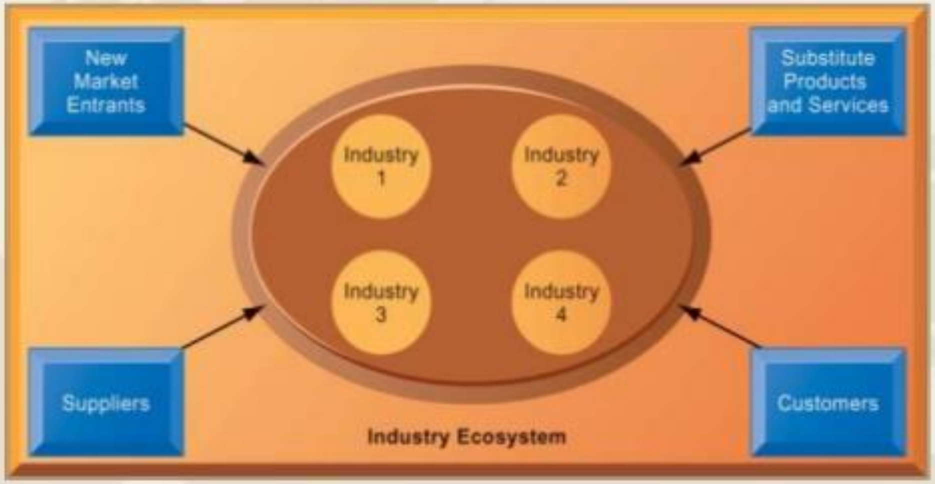


FIGURE 3-13 The digital firm era requires a more dynamic view of the boundaries among industries, firms, customers, and suppliers, with competition occurring among industry sets in a business ecosystem. In the ecosystem model, multiple industries work together to deliver value to the customer. IT plays an important role in enabling a dense network of interactions among the participating firms.

Using Information Systems for Competitive Advantage: Management Issues

- **Sustaining competitive advantage**
  - Because competitors can retaliate and copy strategic systems, competitive advantage is not always sustainable; systems may become tools for survival
- **Performing strategic systems analysis**
  - What is structure of industry?
  - What are value chains for this firm?
- **Managing strategic transitions**
  - Adopting strategic systems requires changes in business goals, relationships with customers and suppliers, and business processes